FrameLens: A Cross-Cultural NLP Framework for Model-Based Literary Interpretation

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https://github.com/LinXinyi0204/nlp-in-industry-FrameLens

Abstract

Large language models (LLMs) can generate sophisticated interpretations of literary texts, but their responses may differ substantially depending on the model and linguistic context. FrameLens investigates these differences through a cross-cultural literary interpretation pipeline that compares responses generated by ChatGPT in English and DeepSeek in Chinese. Rather than treating these differences as inconsistencies to be eliminated, FrameLens makes them explicit and preserves them during synthesis. For each literary question, Claude evaluates the divergence between the two model responses using four framing dimensions derived from Entman's framing theory: problem definition, causal attribution, moral evaluation, and suggested lesson. The identified perspectives are then combined by Claude into an attributed synthesis that explicitly distinguishes model-specific and shared interpretations. We evaluate the system on six literary works using factual, broad, focused, and comparison questions, with synthesis quality assessed through attribution clarity and information coverage. Across twelve synthesized responses, the evaluation achieves an average attribution clarity of 79% and information coverage of 96%. Manual inspection of six additional outputs found no discrepancies with the automated evaluation. These results suggest that explicit attribution can preserve cross-model interpretive differences while retaining the majority of relevant information.

1 Introduction

1.1 Motivation

Large language models (LLMs) have become increasingly capable of interpreting and analyzing literary texts. However, when the same literary question is presented to different models or in different linguistic contexts, the resulting interpretations may emphasize different aspects of the text. Such differences are particularly relevant for literary interpretation, where a single text can support multiple perspectives rather than one objectively correct answer.

In conventional LLM-based analysis, users typically receive a single model response and therefore have limited visibility into alternative interpretations. Differences between models may consequently appear as inconsistencies, even though they can provide valuable insights into how a literary work can be framed from different perspectives. FrameLens addresses this issue by treating interpretive divergence as information rather than noise. The system compares responses generated by ChatGPT in English and DeepSeek in Chinese, identifies differences in their literary framing, and preserves these perspectives during synthesis. This allows users to inspect both shared interpretations and model-specific viewpoints instead of receiving a single merged interpretation.

1.2 Research Aim

The aim of FrameLens is to identify and quantify differences between LLM-generated literary interpretations across models and linguistic contexts, and to preserve these differences in an attributed synthesis. Specifically, the system compares responses generated by ChatGPT in English and DeepSeek in Chinese, evaluates their divergence using Entman's four framing dimensions, and synthesizes the resulting perspectives while retaining their attribution.

2 Methods Compared

2.1 ChatGPT and DeepSeek

FrameLens uses two large language models to generate independent interpretations of the same literary questions. ChatGPT is prompted in English, while DeepSeek is prompted in Chinese. This design allows the system to compare interpretations produced under different model and linguistic contexts, rather than relying on a single model response.

Figure 1 illustrates the core generation process. Given the same literary question, two language-specific prompts are provided independently to ChatGPT and DeepSeek. The resulting responses are kept separate and are only compared in the subsequent evaluation stage.

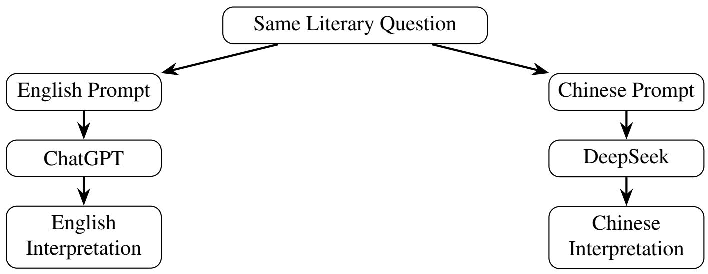

Figure 1: Independent generation of literary interpretations by ChatGPT and DeepSeek under different linguistic contexts.

For a literary question q, the two model responses can be represented as

$$y_{\mathrm{ChatGPT}} = f_{\mathrm{ChatGPT}}(q_{\mathrm{en}}), \qquad y_{\mathrm{DeepSeek}} = f_{\mathrm{DeepSeek}}(q_{\mathrm{zh}}), \tag{1}$$

where q_{en} and q_{zh} denote the English and Chinese versions of the question, respectively, and y_{ChatGPT} and y_{DeepSeek} denote the corresponding literary interpretations. The two responses are generated independently, without exposing either model to the response produced by the other.

The evaluation dataset contains six literary works: *Romeo and Juliet*, *Journey to the West*, *Dream of the Red Chamber*, *Wuthering Heights*, *Don Quixote*, and *The Tale of the Bamboo Cutter*. For each work, four types of questions are considered: factual (*fact*), broad interpretive (*broad*), focused interpretive (*focused*), and comparative (*comparison*). Thus, the total number of question instances is

$$N = W \times Q = 6 \times 4 = 24, \tag{2}$$

where $W = 6$ denotes the number of literary works and $Q = 4$ denotes the number of question types.

The four question types serve different purposes. The *fact* questions primarily test agreement on explicit information contained in the literary works. The *broad* questions require a general interpretation of a work or its central conflict, while the *focused* questions examine the interpretation of a specific character, event, or decision. The *comparison* questions directly ask the models to discuss differences between Chinese and English literary traditions and therefore provide an additional perspective on cross-linguistic interpretation.

The resulting model responses form paired inputs for the subsequent divergence analysis. Importantly, the two models are not asked to reconcile their interpretations themselves. Instead, their responses remain independent until the evaluation and synthesis stages, allowing FrameLens to explicitly identify both shared interpretations and model-specific perspectives.

2.2 Entman-based Divergence Scoring

To compare the interpretations generated by ChatGPT and DeepSeek, FrameLens uses an Entman-inspired framing scheme to characterize their interpretive divergence. Instead of evaluating which response is “correct”, the evaluation focuses on how the two models frame the same literary question differently. A third-party model, Claude, is used as the auditor to compare the two responses independently of the models that generated them.

The comparison is based on four framing dimensions derived from Entman’s framing theory: *Problem Definition* (PD), *Causal Attribution* (CA), *Moral Evaluation* (ME), and *Suggested Lesson* (SL). These dimensions capture four complementary aspects of a literary interpretation: what the central problem or conflict is, what causes it, how the relevant characters or events are morally evaluated, and what broader lesson or implication is derived from the text.

For each dimension, Claude assigns a divergence score between 0 and 3. A score of 0 indicates that the two models adopt essentially the same framing, whereas higher scores indicate increasingly different interpretive frameworks. The scoring scheme is summarized in Table 1.

Table 1: Entman-based divergence scoring scheme.

Score	Interpretation
0	Same core framing or interpretation.
1	Same general framework, but with different emphasis.
2	Different but compatible interpretive frameworks.
3	Fundamentally different interpretive frameworks.

For a given literary question, let d_{PD}, d_{CA}, d_{ME}, and d_{SL} denote the divergence scores assigned to the four dimensions. The overall divergence score is defined as

$$D = d_{\mathrm{PD}} + d_{\mathrm{CA}} + d_{\mathrm{ME}} + d_{\mathrm{SL}}, \qquad 0 \leq D \leq 12. \tag{3}$$

Thus, a higher value of D indicates a greater difference between the framing strategies of the two model responses. The score is not intended to measure the quality or correctness of either interpretation. Instead, it quantifies the extent to which the models approach the same literary question from different interpretive perspectives.

The four dimensions can be understood as follows. *Problem Definition* examines whether the models identify the same underlying conflict or focus on different problems. *Causal Attribution* compares the causes assigned to the conflict, such as individual decisions versus social or institutional factors. *Moral Evaluation* considers whether the models evaluate characters, actions, or events in similar or contrasting ways. Finally, *Suggested Lesson* examines the broader lesson or implication that each model derives from the literary work.

2.3 Attributed Synthesis

After the two model responses have been independently generated and their framing divergence has been evaluated, FrameLens produces an attributed synthesis using Claude. The purpose of this stage is not to select one interpretation as the correct answer or to merge the two responses into a single undifferentiated interpretation. Instead, the synthesis is designed to preserve the perspectives of both models while making their similarities and differences explicit.

The synthesis process takes the two original responses together with the corresponding divergence analysis as input. Each interpretation is retained with an explicit attribution to its source model. When the two models reach a similar interpretation, the synthesis identifies their agreement. When their interpretations differ, both perspectives are presented rather than being arbitrarily reconciled into a single claim.

Formally, let y_{ChatGPT} and y_{DeepSeek} denote the two model responses and let D denote their Entman-based divergence score. The synthesis can be represented as

$$y_{\mathrm{syn}} = g_{\mathrm{Claude}}\left(y_{\mathrm{ChatGPT}}, y_{\mathrm{DeepSeek}}, D\right), \tag{4}$$

where y_{syn} is the resulting attributed synthesis and g_{Claude} denotes the synthesis procedure performed by Claude. The divergence information provides an explicit description of the differences between the two interpretations and helps guide the synthesis towards preserving these differences.

The synthesis follows three main principles. First, model-specific interpretations are explicitly attributed to their source, for example by identifying a perspective as originating from ChatGPT or DeepSeek. Second, shared interpretations are presented as areas of agreement rather than being unnecessarily attributed to only one model. Third, substantial disagreements are presented side by side, allowing the reader to observe different interpretations of the same literary element without forcing an artificial consensus.

This attribution mechanism is particularly important for literary interpretation because divergent answers are not necessarily contradictory or erroneous. Two interpretations may emphasize different causes, moral evaluations, or broader lessons while remaining compatible with the same literary text. FrameLens therefore treats disagreement as an informative part of the output rather than as a problem that must be eliminated.

The resulting synthesis is structured around the major points of agreement and divergence between the two models. It concludes by highlighting the key differences identified during the framing analysis. In this way, the final output provides both a concise interpretation of the literary work and explicit information about how the two model perspectives differ.

3 System Architecture

3.1 Cross-Lingual Model Generation

The first stage of the FrameLens pipeline generates independent responses from ChatGPT and DeepSeek for the same literary work and underlying question. The two models operate in different linguistic settings: ChatGPT receives an English prompt and DeepSeek receives a Chinese prompt. The purpose of this design is to make differences between English and Chinese model-generated interpretations observable while keeping the underlying literary question constant.

The question set contains four types of questions: factual (*fact*), broad interpretive (*broad*), focused interpretive (*focused*), and comparative (*comparison*). Factual questions provide a baseline for examining whether the two models agree on explicit and verifiable information from the literary work. The remaining question types require increasingly interpretive reasoning and therefore provide the main basis for analyzing differences in literary framing.

For a given underlying question q, the system constructs language-specific prompt representations and sends them independently to the two models. The generation process can be formalized as

$$q \xrightarrow{\text{prompting}} (q_{\text{en}}, q_{\text{zh}}) \xrightarrow{f_{\text{models}}} (y_{\text{ChatGPT}}, y_{\text{DeepSeek}}), \tag{5}$$

where q denotes the underlying question, q_{en} and q_{zh} denote its English and Chinese prompt representations, and y_{ChatGPT} and y_{DeepSeek} denote the resulting model responses. The two responses are generated independently, and neither model receives the response produced by the other.

The factual questions serve as a baseline rather than as the main target of the interpretive analysis. They ask about information that can be directly verified from the literary work, such as characters, events, or plot details. Agreement on these questions provides a baseline for examining whether the two responses diverge because of different interpretations or because of disagreement about basic information in the text. Factual consistency does not by itself establish that either response is objectively correct; it is used to identify potential factual disagreement separately from interpretive divergence.

The broad and focused questions instead require the models to interpret the literary work. Broad questions address general themes, central conflicts, or underlying causes, whereas focused questions examine particular characters, events, or decisions. Comparison questions explicitly ask the models to discuss differences between the Chinese and English responses or literary contexts. Together, these question types provide complementary ways of examining how interpretive differences emerge under the two linguistic settings.

The resulting responses are stored as paired outputs together with the corresponding literary work and question type. These paired responses form the input to the subsequent divergence auditing stage, where differences are evaluated using the four Entman-based framing dimensions described in Section 2.2.

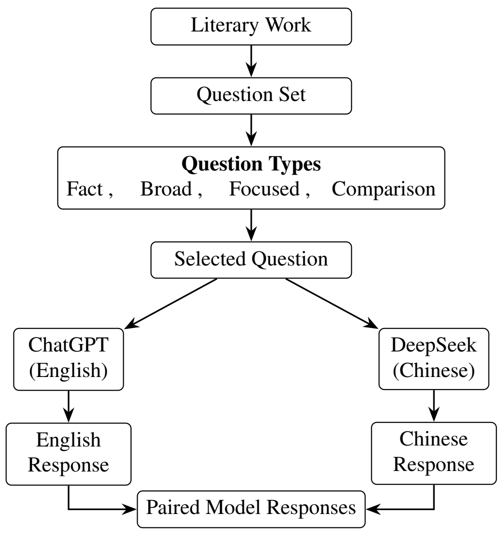

Figure 2: Cross-lingual question and model generation pipeline in FrameLens. Each selected question is independently answered by ChatGPT in English and DeepSeek in Chinese, producing paired responses for subsequent divergence analysis.

3.2 Divergence Auditing

After generating the English and Chinese responses, FrameLens evaluates their interpretive divergence. A third-party language model, Claude, is used as the divergence auditor. It receives the original question and the two independently generated responses, and evaluates their differences using the four framing dimensions defined in Section 2.2.

The auditing process is summarized as

$$(y_{\text{ChatGPT}}, y_{\text{DeepSeek}}, q) \xrightarrow{\text{Claude Audit}} (d_{\text{PD}}, d_{\text{CA}}, d_{\text{ME}}, d_{\text{SL}}, D)\,. \tag{6}$$

Here, q denotes the original question, y_{ChatGPT} and y_{DeepSeek} the English and Chinese responses, respectively, and d_{PD}, d_{CA}, d_{ME}, and d_{SL} the dimension-level divergence scores. D denotes the aggregated divergence score defined in Section 2.2.

In addition to the numerical scores, Claude provides a short dimension-level rationale and assigns an overall qualitative category: *surface*, *moderate*, or *fundamental*. These outputs make the measured divergence interpretable by indicating both its magnitude and the framing dimensions in which the responses differ.

The divergence score measures *interpretive difference*, not response quality. A higher score therefore does not imply that one model is incorrect. Instead, the audit identifies and makes explicit differences between the English and Chinese model-generated interpretations before the attributed synthesis stage.

3.3 Synthesis Pipeline

The final stage of FrameLens transforms the paired model responses into an attributed synthesis. This stage is designed to preserve the differences identified between the English and Chinese interpretations rather than reducing them to a single neutral answer. For this reason, the synthesis is performed by a third-party model, Claude, rather than by either of the two models whose responses are being compared.

The synthesis process is applied to the broad and focused questions for all six literary works. For each question, Claude receives the two original model responses and generates a structured synthesis that explicitly distinguishes shared and model-specific perspectives. The process can be represented as

$$(y_{\text{ChatGPT}}, y_{\text{DeepSeek}}, q) \xrightarrow{\text{Claude}} y_{\text{syn}}, \tag{7}$$

where q denotes the original literary question, y_{ChatGPT} and y_{DeepSeek} denote the English and Chinese model responses, respectively, and y_{syn} denotes the resulting attributed synthesis.

The synthesis follows four main principles. First, model-specific claims are explicitly attributed using labels such as *[ChatGPT]* and *[DeepSeek]*. Second, when both responses express a compatible interpretation, the synthesis explicitly identifies this as an area of agreement rather than assigning the claim to only one model. Third, when the responses differ, both perspectives are presented side by side without forcing an artificial resolution or selecting one interpretation as the correct one. Finally, the synthesis concludes with a concise statement summarizing the key divergence between the two responses.

The resulting synthesis is therefore not intended to produce a single "average" interpretation. Instead, it provides a compact representation of the interpretive space covered by the two responses. A synthesis can preserve a shared observation while simultaneously retaining different explanations, evaluations, or lessons proposed by the two models. This design is particularly important for literary interpretation, where divergent readings can be complementary rather than mutually exclusive.

The synthesis can be viewed as a transformation from two independent perspectives into three types of information:

$$y_{\text{syn}} = \{S_{\text{shared}}, S_{\text{specific}}, D_{\text{key}}\}, \tag{8}$$

Here, S_{shared} denotes shared perspectives, S_{specific} denotes model-specific perspectives, and D_{key} denotes the key divergence between the two responses.

This structure allows the final output to retain the provenance of interpretive claims. Consequently, readers can distinguish what is shared between the English and Chinese responses from what is specific to one model. The synthesis stage therefore complements the numerical divergence audit: while the audit quantifies the magnitude and dimensions of difference, the synthesis explains and presents those differences in a human-readable form.

4 Evaluation

4.1 Experimental Setup

We evaluate FrameLens on six literary works with four questions per work, resulting in 24 paired comparisons. The questions cover four types: *fact*, *broad*, *focused*, and *comparison*. For each question, ChatGPT generates an English response and DeepSeek generates a Chinese response independently. Divergence is then assessed along the four framing dimensions defined in Section 2.2.

The evaluation consists of two parts. First, all 24 paired responses are used for divergence auditing. Second, synthesis is evaluated on the 12 broad and focused cases, with one synthesis generated for each case. The main experimental configuration is summarized in Table 2.

Table 2: Experimental setup of FrameLens.

Component	**Setting**
Literary works	6
Questions per work	4
Total questions	24
Question types	Fact, Broad, Focused, Comparison
English model	ChatGPT
Chinese model	DeepSeek
Response languages	English / Chinese
Divergence dimensions	PD, CA, ME, SL
Score per dimension	0–3
Maximum total score	12
Synthesis cases	12

4.2 Divergence Evaluation

The divergence evaluation measures how differently ChatGPT and DeepSeek frame the same literary question. Importantly, the evaluation does not measure which response is better or more factually correct. Instead, it quantifies differences in interpretation between the English response generated by ChatGPT and the Chinese response generated by DeepSeek.

Following Entman's framing perspective, each pair of responses is examined along four dimensions: Problem Definition, Causal Attribution, Moral Evaluation, and Suggested Lesson. These dimensions capture different stages of interpretation: what the response identifies as the central issue, what it considers responsible for that issue, how it evaluates the relevant actions or events, and what broader conclusion or lesson it derives.

Each dimension is assigned a score from 0 to 3. A score of 0 indicates that the two responses use essentially the same framing for the dimension.

For each question, the four dimension scores are retained separately and also aggregated into an overall divergence score. The resulting score provides a compact measure of the magnitude of the observed difference, while the individual dimensions preserve information about where the difference occurs. This distinction is important because two response pairs can obtain similar total scores while differing in completely different framing dimensions.

The evaluation covers three question types: factual questions, broad interpretive questions, and focused interpretive questions. Factual questions primarily test whether the models converge on explicit information from the literary work. Broad questions ask about the significance, themes, or central conflicts of a work, while focused questions target a particular character, event, or interpretive judgment. This design allows the analysis to distinguish differences arising during factual retrieval from those emerging during literary interpretation.

The resulting scores are analyzed at two levels. First, total divergence scores are compared across literary works and question types to identify general patterns in model disagreement. Second, the four framing dimensions are analyzed separately to determine whether divergence is concentrated in particular aspects of interpretation. This dimension-level analysis is especially important for distinguishing agreement about what happened in a text from disagreement about why it happened, how it should be evaluated, and what meaning should be derived from it.

The evaluation therefore treats divergence as a structured property of the two responses rather than as a single binary outcome. A low score indicates close interpretive alignment, whereas a high score indicates that the two responses construct substantially different readings of the same literary material. The resulting annotations form the basis for the quantitative results presented in Section 5.1 and the qualitative analysis in Section 5.3.

4.3 Synthesis Quality Evaluation

The divergence audit evaluates the differences between the two source responses, but this alone does not establish whether FrameLens can communicate these differences effectively to a reader. We therefore introduce a second evaluation stage for the synthesis output. The goal is to determine whether the final synthesis preserves the relevant information from both responses and whether individual claims remain traceable to their sources.

We evaluate the resulting texts using two complementary metrics: *Attribution Clarity* and *Information Coverage*. Attribution Clarity measures whether readers can identify which source model supports a particular interpretive claim. This metric is central to FrameLens because the purpose of the synthesis is not simply to generate a fluent new interpretation, but to preserve the provenance of the different perspectives.

Information Coverage measures how much of the relevant content from the two original responses is retained in the synthesis. A high coverage score indicates that the synthesis preserves the main interpretive arguments, while a lower score indicates that some information from the source responses has been omitted. Coverage is therefore interpreted together with the nature of the omitted information: losing a secondary example is less consequential than losing a central interpretive claim.

The two metrics capture complementary aspects of synthesis quality. Attribution Clarity evaluates whether the origin of an interpretation remains visible, whereas Information Coverage evaluates whether the content itself is preserved. A synthesis could therefore have high coverage but poor attribution, or clear attribution while omitting important source information. Evaluating both properties is necessary to assess whether the synthesis fulfills the intended function of FrameLens.

To provide an additional reliability check, the automated evaluation is complemented by manual inspection. Six complete synthesis outputs from three literary works—*Romeo and Juliet*, *Dream of the Red Chamber*, and *Don Quixote*—are read manually and compared with the automated assessment. This cross-check is intended to determine whether the numerical evaluation is consistent with human inspection of the actual synthesis texts.

The synthesis evaluation therefore addresses a different question from the divergence audit. The divergence audit asks whether the two source responses differ and where those differences occur. The synthesis evaluation asks whether FrameLens can preserve and communicate these differences clearly without losing the main information. The results of this evaluation are reported in Section 5.2, while the complete case-level results are provided in Appendix.

5 Results and Discussion

5.1 Divergence Results

Table 3 summarizes the overall divergence scores across the six literary works and three question types evaluated in the main divergence analysis. The results show a clear difference between factual and interpretive questions. Fact questions produce consistently low divergence scores, ranging from 0 to 2, whereas broad and focused questions yield substantially higher scores, ranging from 5 to 9.

Table 3: Divergence scores by literary work and question type.

Literary Work	Fact	Broad	Focused	Highest dimension
Romeo and Juliet	2	7	6	CA / ME / SL
Journey to the West	0	7	5	PD / CA / SL
Dream of the Red Chamber	0	7	9	ME
Wuthering Heights	0	7	8	PD / CA / ME / SL
Don Quixote	2	6	8	PD / CA / ME / SL
The Tale of the Bamboo Cutter	0	5	5	SL

The divergence analysis reveals a clear difference between factual and interpretive questions. Across the six literary works, factual questions show consistently low divergence: four of the six works receive a total score of 0, while the remaining two receive a score of 2. In contrast, broad and focused interpretive questions produce substantially higher divergence scores. This indicates that the models generally converge on explicit factual information but diverge more when interpretation, evaluation, and broader meaning are required.

An additional observation is that the observed divergence is consistent with the models' own assessment of their differences. Before the detailed divergence audit, the models were asked to identify potential differences between their respective interpretations. The patterns identified in these self-assessments largely correspond to the differences found by the subsequent structured evaluation. This agreement suggests that the observed model bias is not merely an accidental finding in a small number of examples, but reflects a more systematic difference in how the two models construct literary interpretations.

The divergence scores also vary across framing dimensions. Problem Definition generally shows lower divergence, whereas Moral Evaluation and Suggested Lesson tend to show stronger differences for interpretive questions. This suggests that the models may agree more often on what a literary situation concerns while differing more substantially in how it should be evaluated and what broader meaning should be derived from it.

5.2 Synthesis Quality Results

The synthesis evaluation shows that FrameLens is generally effective at preserving the two model perspectives while keeping their sources traceable. Across the 12 synthesis cases, attribution clarity averages 79%, with individual scores ranging from 75% to 83%. Information coverage is substantially higher, averaging 96% with a range of 80%–100%.

Table 4: Summary of synthesis quality across 12 cases.

Metric	Mean	Range
Attribution Clarity	79%	75–83%
Information Coverage	96%	80–100%

The high information coverage indicates that the synthesis generally retains the core arguments from both source responses. The main information losses concern secondary or low-density details rather than central interpretations. For example, in the focused *Dream of the Red Chamber* case, the symbolic meaning of the cloak was omitted, while the central disagreement concerning moral responsibility was preserved.

Attribution clarity is lower than information coverage, although this difference is partly explained by the synthesis format. Some statements were intentionally presented as shared observations when ChatGPT and DeepSeek expressed similar interpretations. Consequently, the 79% attribution score should not be interpreted as an equivalent percentage of incorrect attribution.

The manual review of six complete synthesis outputs was consistent with the automated GPT-4o evaluation, with no mismatches reported. This provides additional support for the reliability of the automatic assessment. Overall, the results suggest that FrameLens can preserve the main information from both model responses while presenting their respective perspectives in an explicitly attributed form.

The complete case-level evaluation, including attribution clarity and information coverage for all 12 synthesis outputs, is provided in Appendix Table 7. The appendix also reports the corresponding literary work and question type for each case, allowing the aggregate results presented above to be traced back to the individual evaluations.

5.3 Qualitative Analysis

The quantitative divergence scores reveal a clear pattern across the four framing dimensions. Table 5 summarizes the mean scores for factual, broad, and focused questions.

Table 5: Mean divergence scores across framing dimensions and question types.

Dimension	Fact	Broad	Focused
Problem Definition	0.0	1.5	1.5
Causal Attribution	0.2	1.7	1.5
Moral Evaluation	0.3	1.3	1.8
Suggested Lesson	0.5	2.0	1.8

The results show that factual questions produce consistently low divergence across all four dimensions. For interpretive questions, divergence increases substantially. Problem Definition remains relatively stable at 1.5 for both broad and focused questions, whereas Suggested Lesson reaches the highest mean divergence for broad questions (2.0). For focused questions, Moral Evaluation and Suggested Lesson show the highest mean divergence, both reaching 1.8. These results suggest that the two model responses tend to differ less in identifying the central issue than in evaluating its meaning and deriving broader implications.

To illustrate this pattern in more detail, we examine the strongest individual case: the focused question on *Dream of the Red Chamber*. This case received a total divergence score of 9/12 and was the only case classified as *fundamental* divergence. Its dimension-level scores are shown in Table 6.

Table 6: Divergence scores for the strongest observed case.

Dimension	Score
Problem Definition	2
Causal Attribution	2
Moral Evaluation	3
Suggested Lesson	2
Total Divergence	**9/12**
Divergence Type	*Fundamental*

The question concerns Baoyu's acceptance of Qingwen's request to repair his cloak while she is seriously ill. ChatGPT interprets the action primarily through Baoyu's privilege, social hierarchy, and insufficient awareness of Qingwen's suffering. DeepSeek, in contrast, interprets the same action as a conscious and compassionate response that respects Qingwen's emotional agency. Thus, the two responses differ not only in wording but also in how the same literary event is framed and evaluated. The synthesis preserves this contrast by explicitly attributing the two interpretations to the respective models.

The strongest difference occurs in Moral Evaluation, which receives the maximum score of 3. ChatGPT gives the action a mildly critical interpretation, whereas DeepSeek rejects a straightforward moral condemnation and presents the action as a form of emotional understanding. The other three dimensions each receive a score of 2. This case therefore illustrates how the dimension-level analysis complements the overall divergence score: the total score identifies a strong divergence, while the individual dimensions reveal that the most pronounced difference concerns the moral evaluation of the same event.

The comparison questions provide additional evidence for this pattern: the differences anticipated by the models are generally consistent with the divergences identified by the structured audit. However, the self-assessments are not fully reliable. In some cases, the predicted direction of divergence does not match the observed direction, while in others the models identify the correct direction but describe the difference too vaguely. The models may therefore either overestimate or underestimate the actual divergence. In both cases, the structured audit provides a more precise characterization of the observed differences. Therefore, model self-diagnosis should be treated as a supporting signal rather than the primary basis for evaluation. This further motivates the use of a structured, dimension-level audit to obtain a more reliable and consistent assessment of divergence.

5.4 Limitations

Despite the promising results, several limitations should be considered when interpreting the findings.

First, the evaluation is relatively small in scale. The experiments cover six literary works and a limited set of questions, with the main quantitative analysis focusing on factual, broad, and focused questions. Although the selected works span different literary traditions, this sample is not sufficient to establish that the observed divergence patterns generalize to literary interpretation more broadly.

Second, the experimental design does not fully separate model effects from language effects. ChatGPT is evaluated through English responses, whereas DeepSeek is evaluated through Chinese responses. Therefore, the observed differences may reflect a combination of model-specific behavior, language-conditioned generation, and other differences between the two systems. The experiment consequently cannot determine which factor is responsible for an individual divergence.

Third, the results should not be interpreted as direct evidence of Western–Chinese cultural differences. FrameLens compares an English response generated by ChatGPT with a Chinese response generated by DeepSeek. While cultural or linguistic factors may contribute to the observed differences, establishing such a causal relationship would require a controlled design in which model and language are independently varied.

Fourth, the synthesis quality evaluation introduces a potential evaluator bias. The synthesis outputs are evaluated by ChatGPT, while ChatGPT is also one of the two source models whose interpretations are being compared. In particular, the Information Coverage score may be somewhat biased in favor of the ChatGPT response: ChatGPT may judge the retention of its own interpretive content more leniently than the retention of content originating from DeepSeek. Consequently, the reported 96% average Information Coverage should not be interpreted as a completely model-independent measure of synthesis completeness.

Finally, the synthesis process involves an unavoidable trade-off between conciseness and information preservation. High information coverage does not necessarily mean that every detail from both source responses is retained. In addition, the synthesis itself does not encode the numerical strength of a disagreement: readers can identify the two perspectives, but cannot determine from the synthesis alone whether the underlying divergence score is 5 or 9. The divergence audit and the synthesis therefore provide complementary information rather than replacing one another.

6 Conclusion

FrameLens presents a cross-lingual model synthesis pipeline for making differences between LLM-based literary interpretations explicit and traceable. Rather than treating different model responses as inconsistencies that should be resolved, the system preserves their interpretive differences and organizes them through a structured divergence audit and an attributed synthesis. The approach compares an English response generated by ChatGPT with a Chinese response generated by DeepSeek and analyzes their framing differences across Problem Definition, Causal Attribution, Moral Evaluation, and Suggested Lesson.

The evaluation demonstrates that divergence is strongly associated with the interpretive nature of the question. Factual questions show consistently low divergence, whereas broad and focused interpretive questions produce substantially larger differences. In particular, Suggested Lesson and Moral Evaluation tend to exhibit stronger divergence than Problem Definition. The qualitative analysis further shows that high divergence can arise when the two models assign different causes, values, or implications to the same literary event.

The synthesis evaluation indicates that FrameLens can preserve these differences while maintaining a high level of information coverage. Across 12 synthesis cases, the system achieves an average attribution clarity of 79% and information coverage of 96%. Manual inspection of six complete synthesis outputs was consistent with the automated evaluation. These results suggest that the synthesis stage can retain the main interpretive content while making model-specific perspectives visible to the reader.

At the same time, the findings should be interpreted within the scope of the experimental design. In particular, the current setup cannot separate model effects from language effects and does not establish a causal relationship between the observed differences and cultural factors. Future work could therefore extend the evaluation to more literary works, models, languages, and question types, while introducing a controlled design that independently varies model and language.

Overall, FrameLens demonstrates that cross-model disagreement can be treated as useful interpretive information rather than noise. By combining quantitative divergence measurement with source-attributed synthesis, the system provides a transparent way to inspect multiple LLM perspectives on the same literary text.

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Appendix

Table 7: Synthesis quality evaluation results by literary work and question type.

Literary Work	Question Type	Attribution Clarity	ChatGPT Coverage	DeepSeek Coverage	Total Coverage
Romeo and Juliet	broad	75%	100%	100%	100%
Romeo and Juliet	focused	80%	100%	100%	100%
Journey to the West	broad	82%	100%	100%	100%
Journey to the West	focused	83%	100%	100%	100%
Dream of the Red Chamber	broad	83%	100%	100%	100%
Dream of the Red Chamber	focused	80%	100%	100%	90%
Wuthering Heights	broad	83%	75%	100%	88%
Wuthering Heights	focused	75%	80%	80%	80%
Don Quixote	broad	75%	100%	100%	100%
Don Quixote	focused	80%	100%	100%	100%
Bamboo Cutter	broad	75%	100%	100%	100%
Bamboo Cutter	focused	82%	100%	100%	100%

Table 8: Member contributions.

Member	Contribution
Xinyi Lin	• Designed and selected the literary works and questions. • Designed the Control Group questions. • Conducted repeated manual verification of the Entman scoring results, identifying and correcting several scoring issues. • Wrote the in-depth case study analyses for the three core cases and cross-checked them against Control Group B data. • Manually verified the synthesis quality of the three core cases.
Jiaran Li	• Built the full pipeline integrating GPT-4o, DeepSeek, and Claude APIs for query, audit, and synthesis. • Implemented Entman-based framing divergence scoring with Claude as third-party auditor. • Developed automated evaluation of synthesis quality (attribution clarity and coverage). • Built the FrameLens web interface using Streamlit. • Created video and uploaded it to YouTube.
Yiming Li	• Proposed the core project idea and defined the overall research direction. • Designed the initial methodology and system architecture. • Designed the overall framework for cross-lingual model comparison and synthesis. • Wrote the technical report.